

Supplementary material

Table S1. Area of CPIS-equipped fields in countries in arid areas.

Country	Area of CPIS-equipped fields (ha)		
	2021	2000	Increase
Algeria	40600	9209	31391
Angola	1561	133	1428
Argentina	135145	53501	81644
Australia	180252	64601	115651
Azerbaijan	42819	98	42721
Bolivia	4440	1290	3150
Botswana	4437	1202	3235
Brazil	31876	18491	13385
Bulgaria	1966	1511	455
Burkina Faso	2455	2265	190
Canada	358610	250912	107698
Chad	3555	3382	173
Chile	5641	2675	2967
China	192485	3178	189308
Dominican Republic	385	0	385
Ecuador	71	0	71
Egypt	206821	15712	191109
Ethiopia	706	0	706
France	197	123	74
Georgia	426	0	426
Ghana	187	0	187
Honduras	49	0	49
Hungary	946	219	727
India	698	0	698
Iran	18582	3121	15460
Iraq	131192	31	131161
Israel	7623	3402	4222
Italy	335	87	248
Jordan	2801	2560	241
Kazakhstan	35337	7371	27966
Kenya	3035	209	2826
Kuwait	1172	506	666
Kyrgyzstan	581	553	28
Lesotho	181	67	114
Libya	47618	31985	15633
Madagascar	2491	2277	213
Malawi	2835	0	2835
Mali	1556	0	1556
Mauritania	425	0	425
Mexico	334935	36972	297963

Moldova	1654	131	1522
Mongolia	5446	185	5261
Morocco	7143	6992	151
Mozambique	3444	465	2980
Namibia	7263	1209	6054
New Zealand	2716	0	2716
Nigeria	58	0	58
Oman	5839	1276	4563
Pakistan	632	0	632
Paraguay	2375	0	2375
Peru	13638	972	12666
Portugal	37971	18661	19310
Qatar	2617	829	1789
Romania	6767	15	6752
Russia	93339	28245	65095
Saudi Arabia	907558	538433	369124
Senegal	2937	79	2858
Serbia and Montenegro	1485	170	1314
Slovakia	151	0	151
South Africa	563047	213177	349870
Spain	101997	74848	27149
Sudan	70099	0	70099
Swaziland	7579	1389	6190
Syria	6432	1292	5139
Tajikistan	603	0	603
The Gambia	300	0	300
Tunisia	872	300	572
Turkey	37672	0	37672
Turkmenistan	1078	277	801
Ukraine	176557	110515	66042
United Arab Emirates	2535	66	2470
United States	6267677	4308613	1959064
Yemen	128	12	115
Zambia	57413	6666	50746

Table S2. Area of CPIS-equipped fields in the 25 largest aquifers in arid areas.

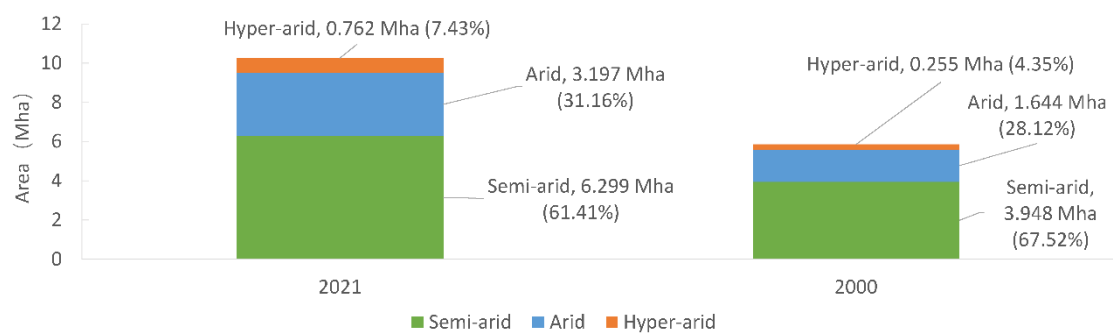
Aquifer	Area of CPIS-equipped fields (ha)		
	2021	2000	Increase
Arabian Aquifer System	911963	541699	370264
Atlantic and Gulf Coastal Plain Aquifers	44240	30353	13887
California Central Valley Aquifer System	7347	2910	4437
Canning Basin	2464	56	2408
East European Aquifer System	42445	7345	35100
Great Artesian Basin	27285	10412	16873
Guarani Aquifer System	9755	3692	6063
Indus Basin	632	0	632
Karoo Basin	280482	108590	171892
Lake Chad Basin	3555	3382	173
Lower Kalahari-Stampriet Basin	619	424	195
Maranhao Basin	71	0	71
Murzuk-Djado Basin	33534	11222	22312
North Caucasus Basin	31844	12487	19357
North China Aquifer System	44	0	44
Northern Great Plains Aquifer	532730	337356	195374
Nubian Aquifer System	235503	35535	199968
Ogallala Aquifer	4052968	3161165	891803
North Western Sahara Aquifer System	38942	9128	29814
Senegalo-Mauritanian Basin	3500	79	3421
Song-Liao Plain	5341	0	5341
Taoudeni-Tanezrouft Basin	18	0	18
Tarim Basin	1103	0	1103
Upper Kalahari-Cuvélai-Upper Zambezi Basin	3587	810	2777
West Siberia Artesian Basin	16285	1811	14474

Table S3. The area of CPIS-equipped fields within 50 km to cities that have populations larger than 500,000.

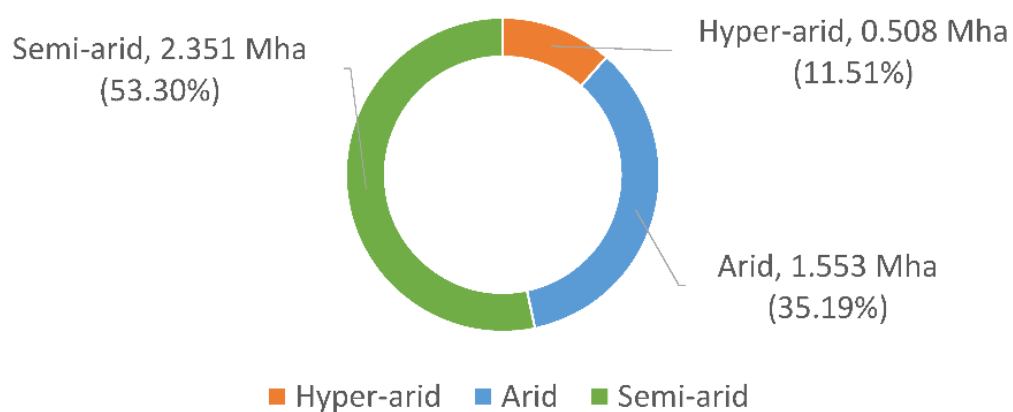
City (Country)	CPIS Area (ha)	City (Country)	CPIS Area (ha)
Abu Dhabi (United Arab Emirates)	75	Madrid (Spain)	381
Ad Damman (Saudi Arabia)	36	Maputo (Mozambique)	25
Addis Ababa (Ethiopia)	78	Marrakech (Morocco)	623
Adelaide (Australia)	681	Mashhad (Iran)	50
Aguascalientes (Mexico)	6790	Mecca (Saudi Arabia)	998
Al Basrah (Iraq)	27	Melbourne (Australia)	198
Al Mawsil (Iraq)	1422	Mendoza (Argentina)	83
Alexandria (Egypt)	512	Merida (Mexico)	1577
Amman (Jordan)	127	Monterrey (Mexico)	126
An Najaf (Iraq)	1563	Odesa (Ukraine)	1275
Ankara (Turkey)	38	Oklahoma City (United States)	2394
Arak (Iran)	621	Omsk (Russia)	612
Arbil (Iraq)	5254	Oran (Algeria)	121
Astana (Kazakhstan)	975	Perth (Australia)	37
Az Zarqa' (Jordan)	152	Phoenix (United States)	1042
Baghdad (Iraq)	2944	Port Elizabeth (South Africa)	1582
Baotou (China)	2727	Port Said (Egypt)	39
Barnaul (Russia)	25	Pretoria (South Africa)	6054
Bishkek (Kyrgyzstan)	638	Puebla (Mexico)	698
Blantyre (Malawi)	1369	Qiqihar (China)	136
Brisbane (Australia)	27	Queretaro (Mexico)	158
Bulawayo (Zimbabwe)	59	Riyadh (Saudi Arabia)	1216
Cairo (Egypt)	184	Rostov-on-Don (Russia)	205
Calgary (Canada)	2211	Sacramento (United States)	160
Cape Town (South Africa)	635	Saltillo (Mexico)	4336
Casablanca (Morocco)	207	San Antonio (United States)	4233
Chihuahua (Mexico)	4326	San Juan (Argentina)	89
Cordoba (Argentina)	5414	San Luis Potosi (Mexico)	167
Culiacan (Mexico)	271	San Miguel De Tucuman (Argentina)	426
Denver (United States)	300	Santiago (Chile)	497
Dnipropetrovs'k (Ukraine)	36	Saratov (Russia)	6380
Doha (Qatar)	2059	Seville (Spain)	1318
Donets'k (Ukraine)	129	Sharjah (United Arab Emirates)	96
Dubai (United Arab Emirates)	462	Suez (Egypt)	549
El Paso (United States)	453	Tabriz (Iran)	74
Esfahan (Iran)	123	Tabuk (Saudi Arabia)	37796
Fes (Morocco)	487	T'bilisi (Georgia)	426
Guadalajara (Mexico)	32	Tegucigalpa (Honduras)	49
Harare (Zimbabwe)	8028	Tehran (Iran)	97
Hermosillo (Mexico)	163	Thies (Senegal)	47
Huhot (China)	916	Tlaxcala (Mexico)	928
Irbid (Jordan)	2414	Tripoli (Libya)	33
Jerusalem (Israel)	112	Tunis (Tunisia)	311
Johannesburg (South Africa)	9864	Ul'yanovsk (Russia)	47
Khartoum (Sudan)	4918	Volgograd (Russia)	3516
Konya (Turkey)	839	Yinchuan (China)	769
Lisbon (Portugal)	4801	Zaporizhzhya (Ukraine)	898
Luanda (Angola)	350	Zaragoza (Spain)	4992
Lusaka (Zambia)	21099		



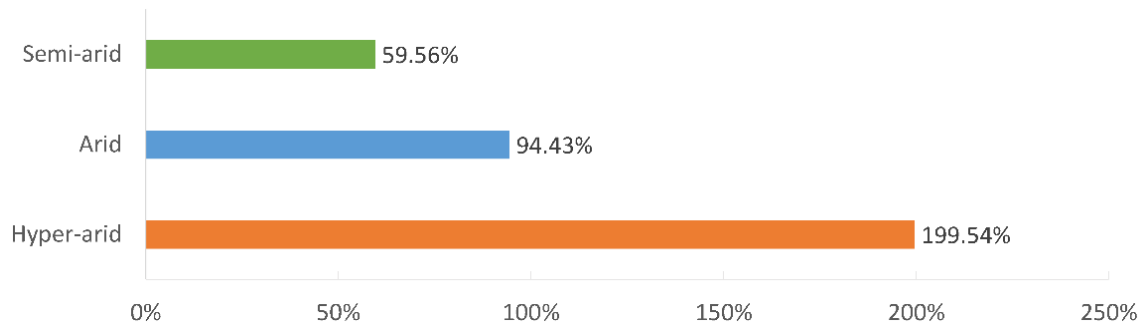
Fig. S1 CPIS samples.



(a) area in Mha and as percentage

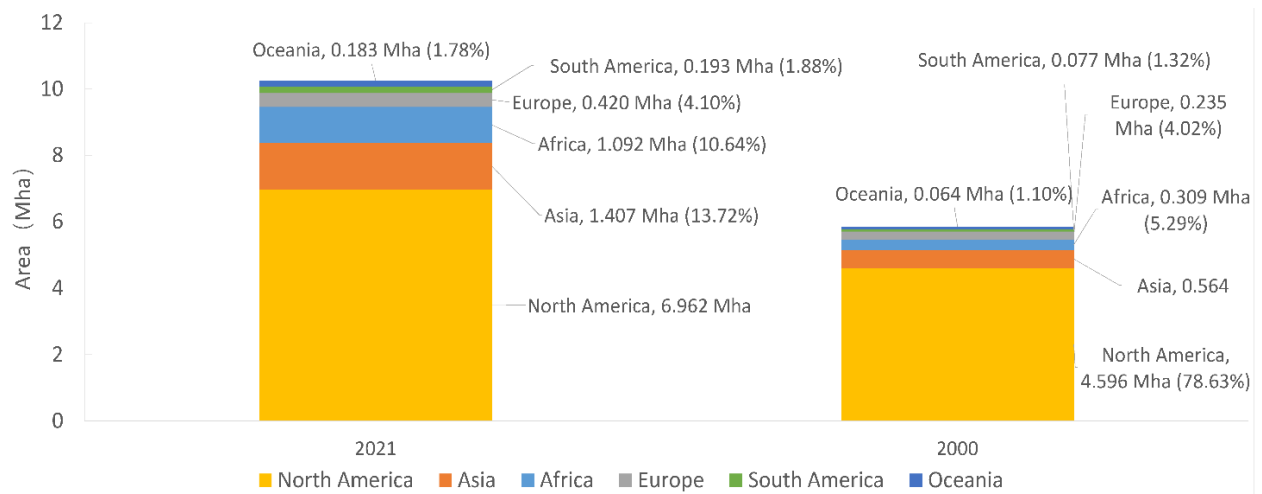


(b) absolute increase in Mha and as percentage

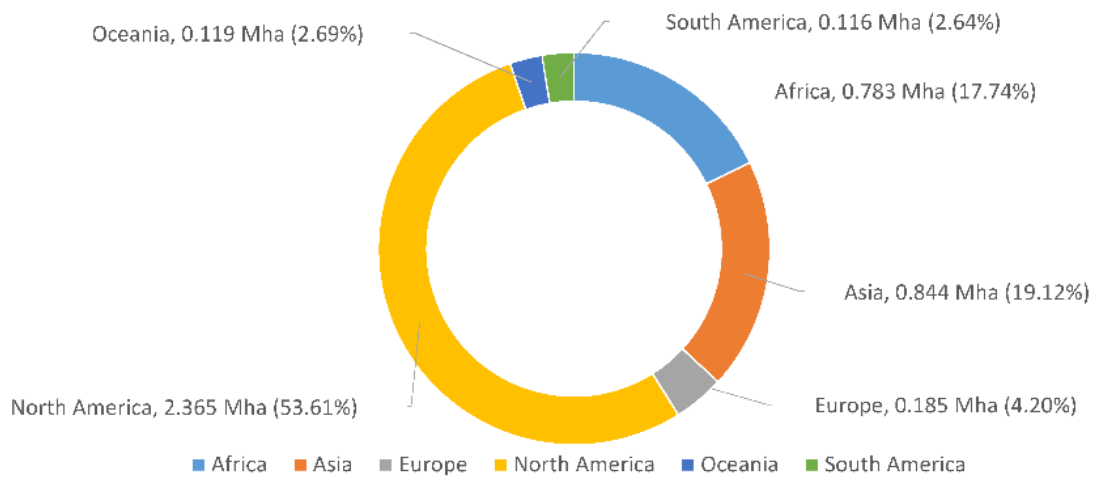


(c) relative increase

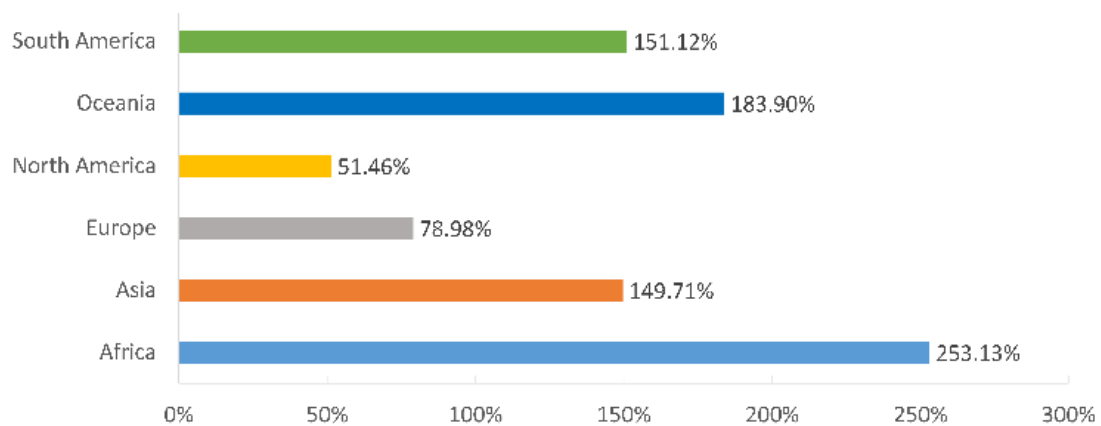
Fig. S2 CPIS at aridity scale. (a) area; (b) absolute increase; (c) relative increase.



(a) area in Mha and as percentage

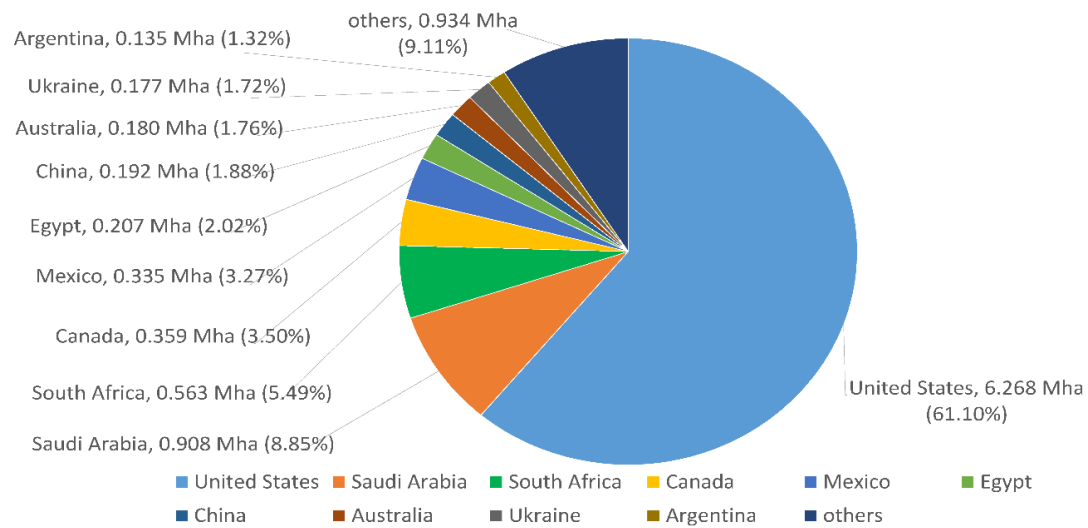


(b) absolute in Mha and percentage increase

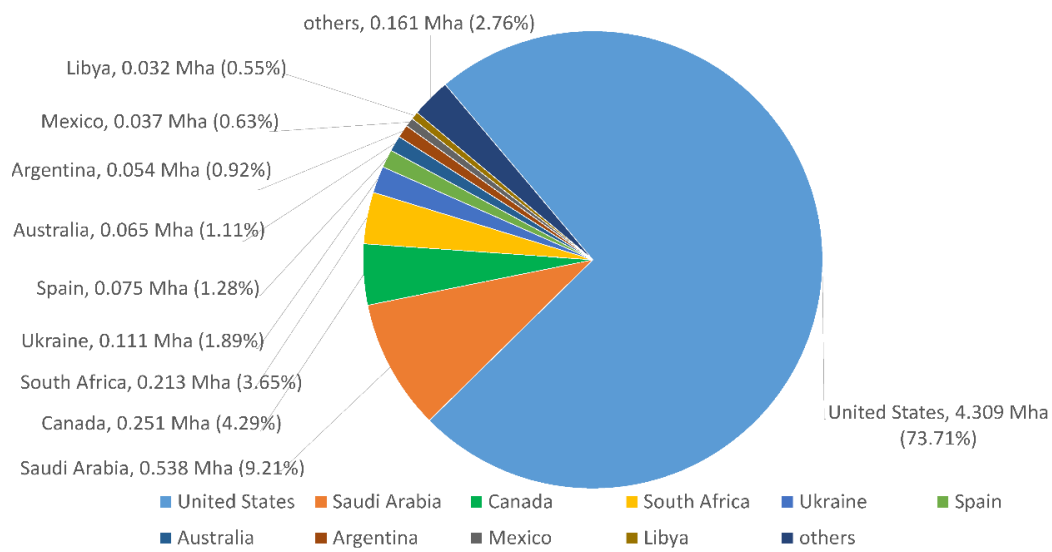


(c) relative increase

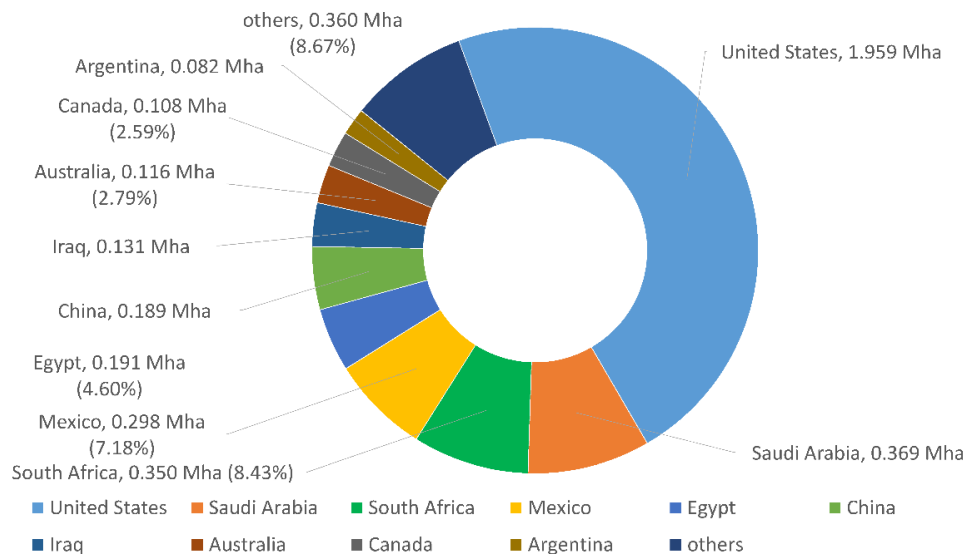
Fig. S3 CPIS at continental scale. (a) area; (b) absolute increase; (c) relative increase.



(a) area in Mha and percentage in 2021



(b) area in Mha and percentage in 2000



(c) absolute in Mha and percentage increase

Fig. S4 CPIS at country scale. (a) top ten countries that have the largest CPIS area in 2021; (b) top ten countries that have the largest CPIS area in 2000; (c) top ten countries that have the largest increase of CPIS area in the past 20 years.

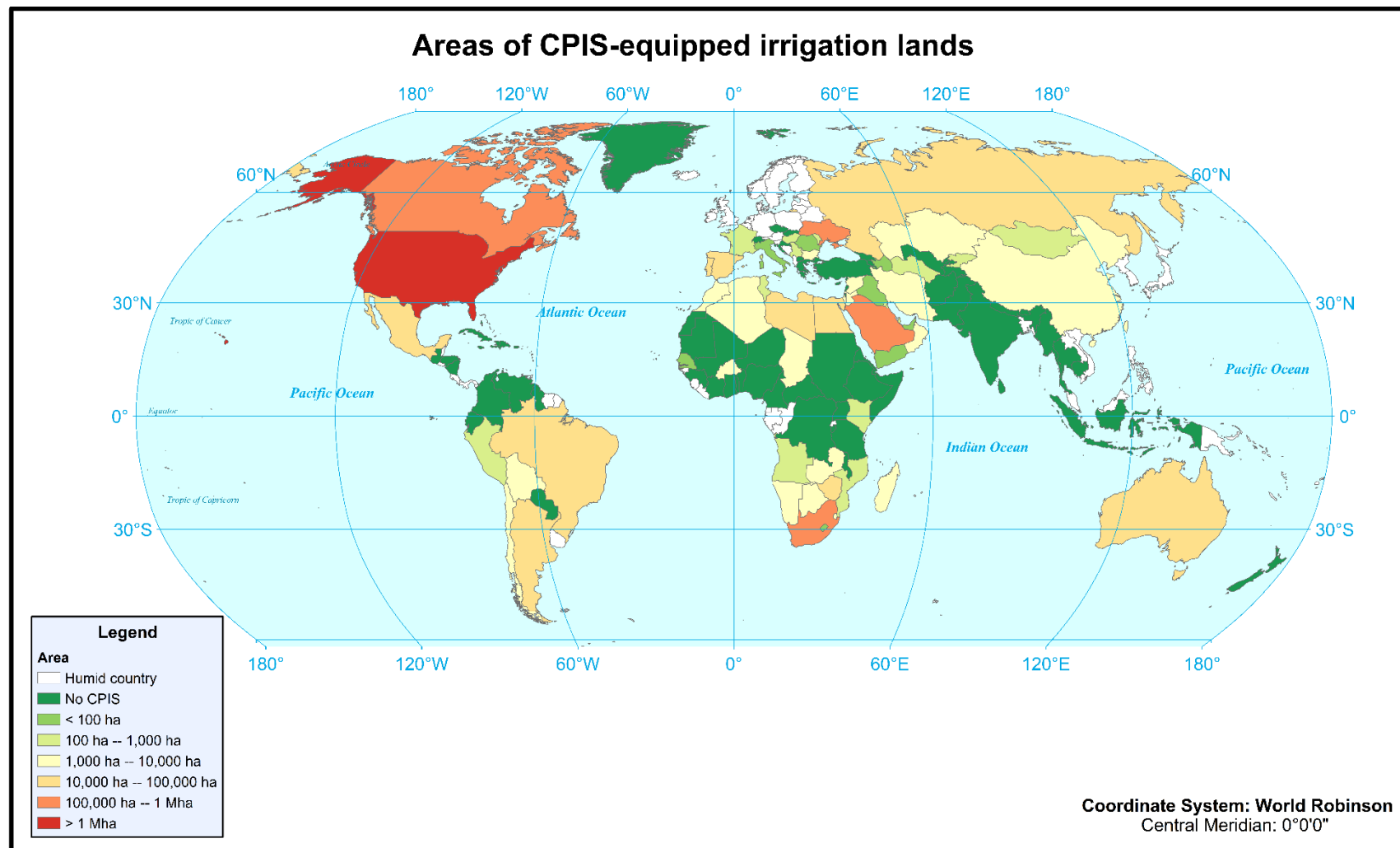
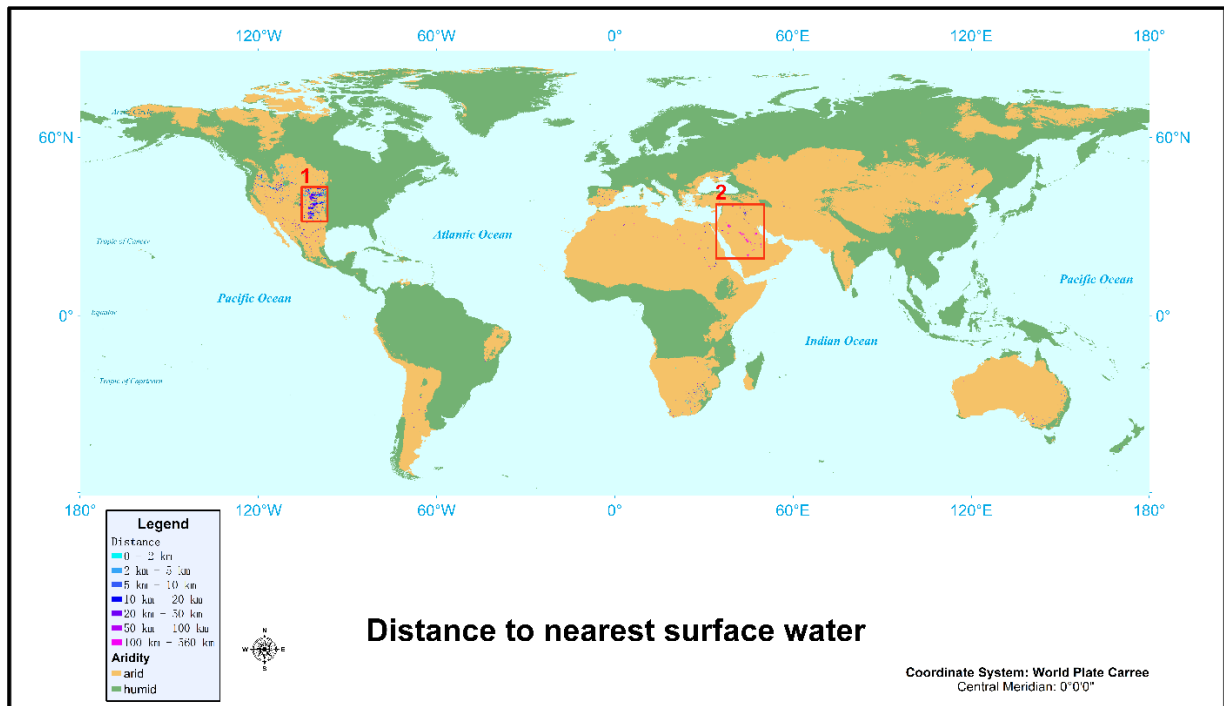
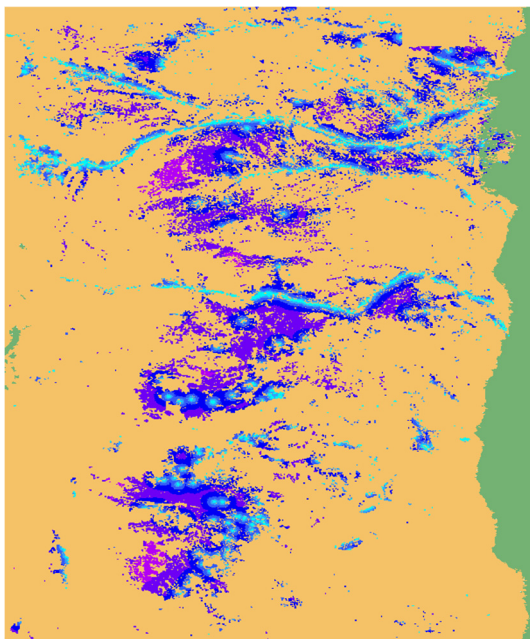


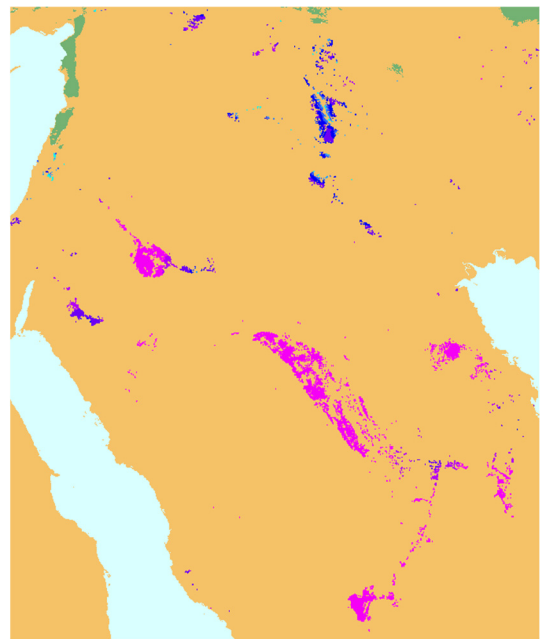
Fig. S5 Area of CPIS-equipped irrigation lands at the country scale in 2000.



(a)



(b)



(c)

Fig. S6 (a) Distance from CPIS to their nearest surface water bodies; (b) zoomed-in of area 1; (c) zoomed-in of area 2.

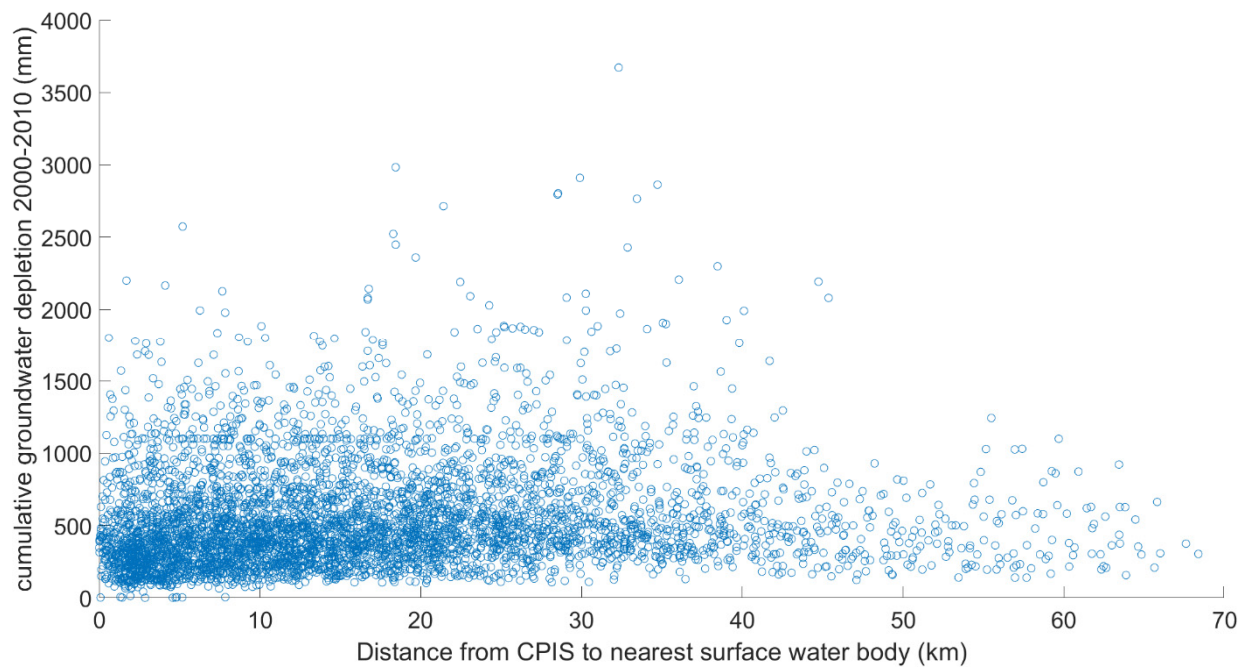


Fig. S7 Scatter plot of correlation between the distance from CPIS to their nearest surface water bodies with the cumulative groundwater depletion data in the Ogallala aquifer (de Graaf et al., 2017, 2019).

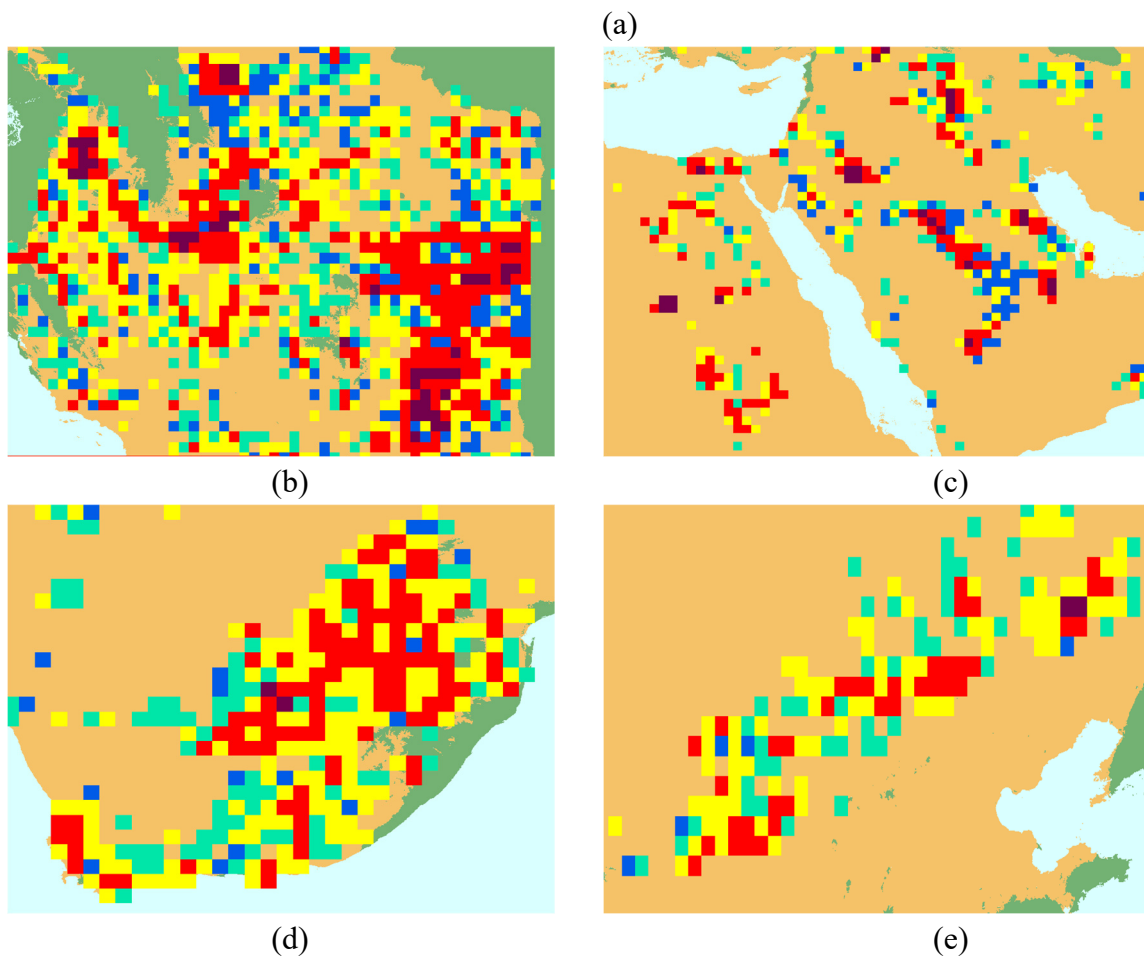
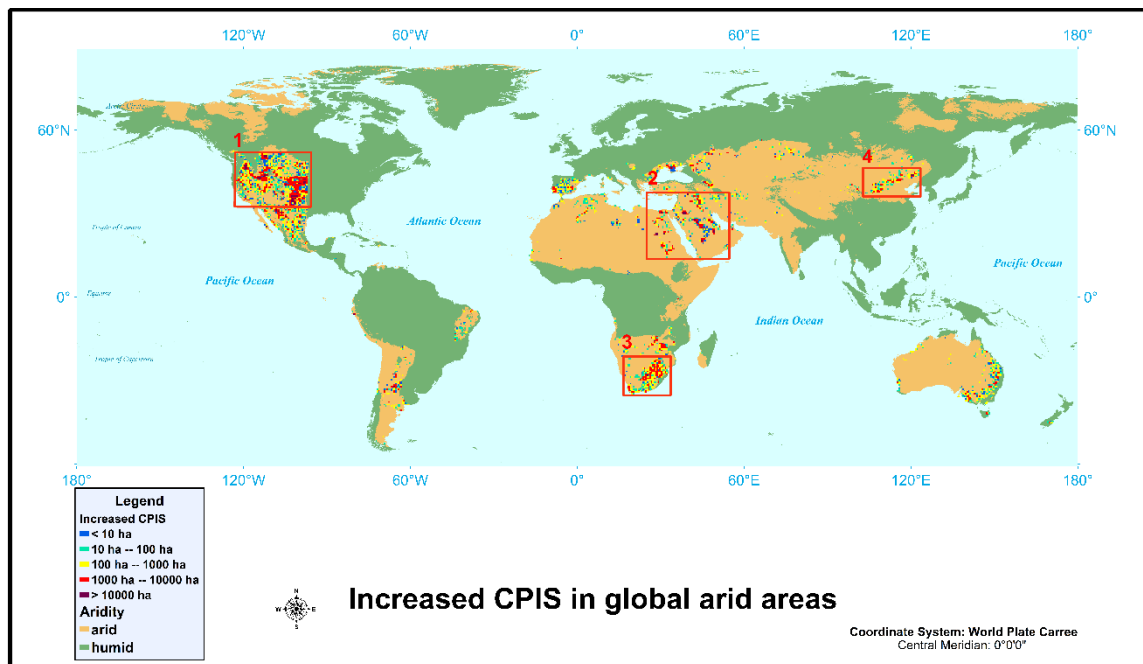


Fig. S8 (a) Increased CPIS area in global arid regions from 2000 to 2021 at a spatial resolution of 0.5 arc-degree; (b) zoomed-in of area 1; (c) zoomed-in of area 2; (d) zoomed-in of area 3; (e) zoomed-in of area 4.

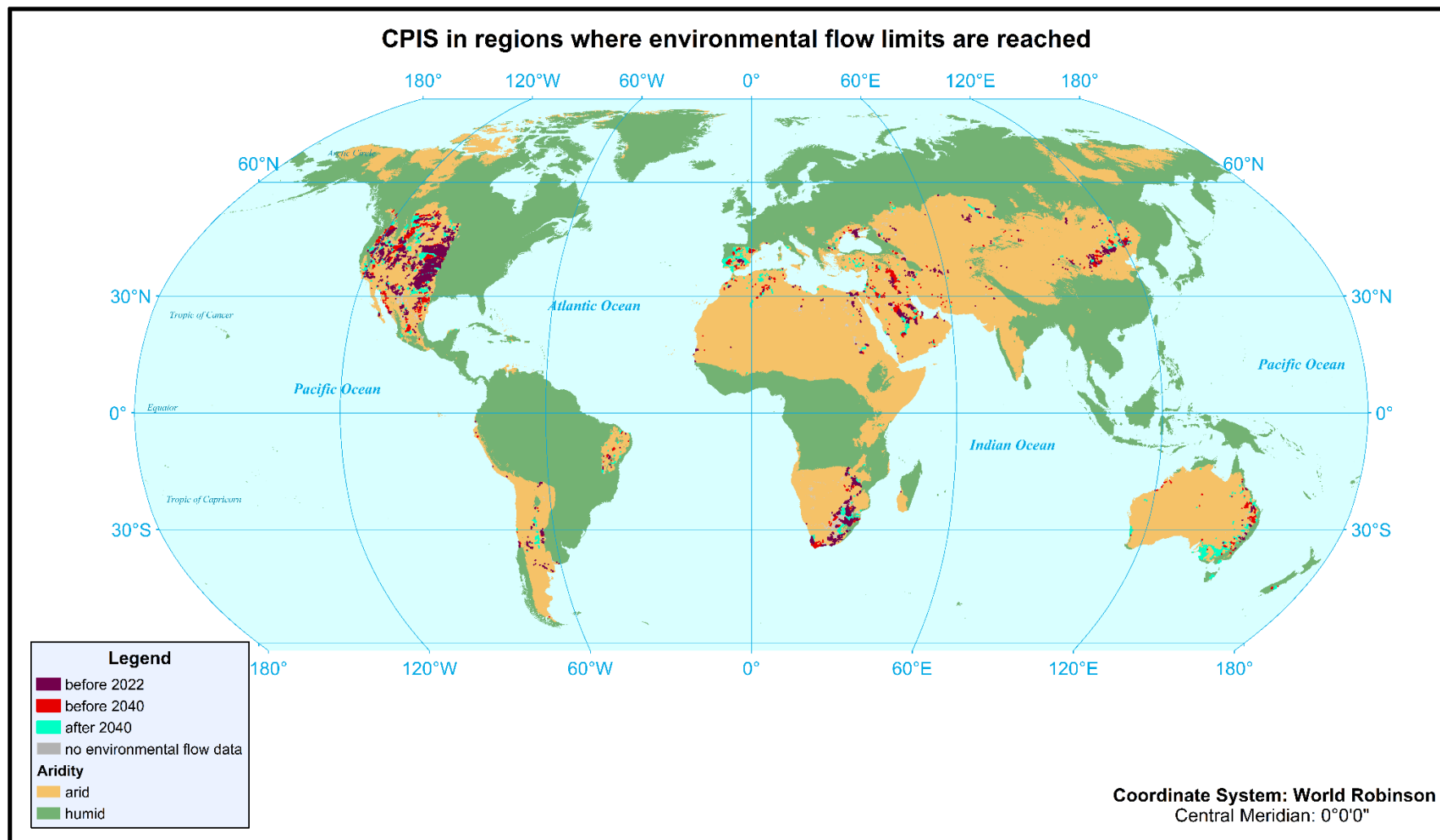


Fig. S9 CPIS in regions where environmental flow limits are reached.

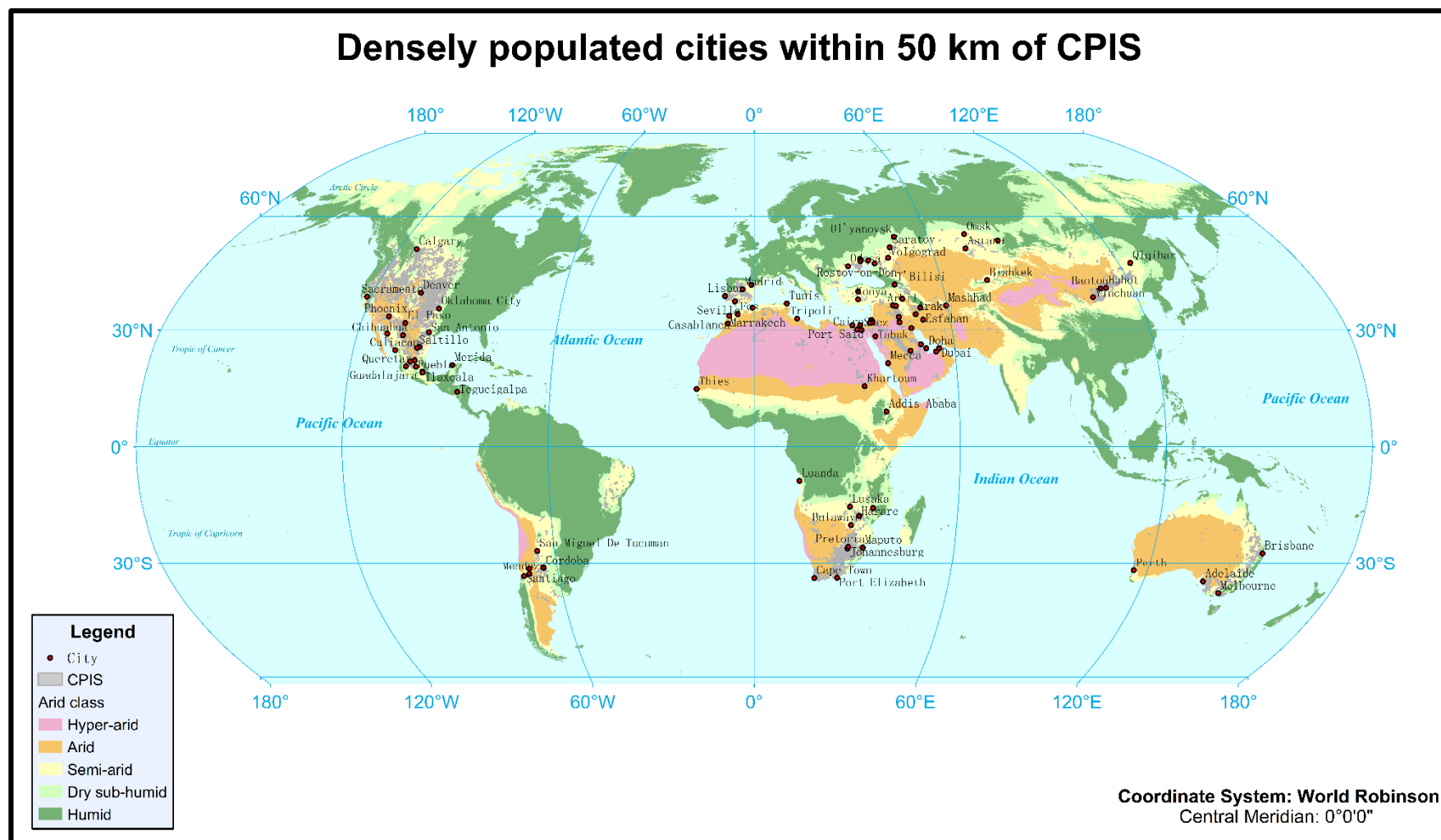
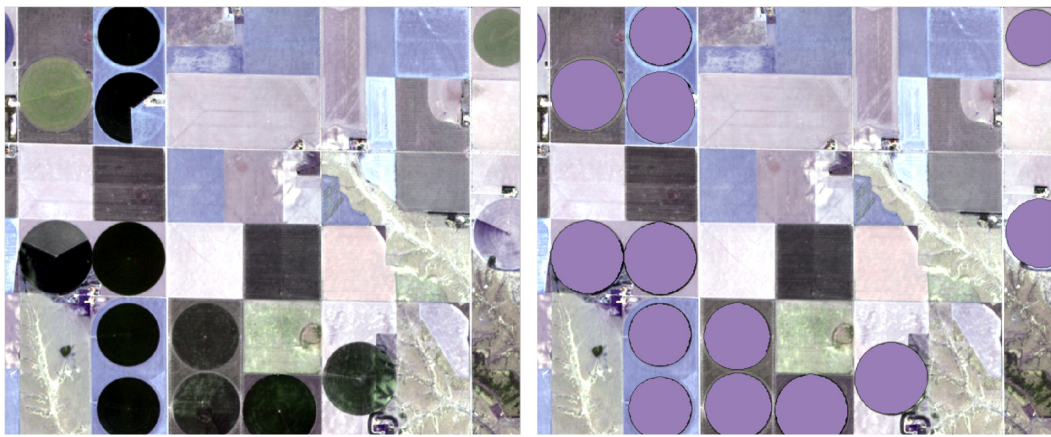
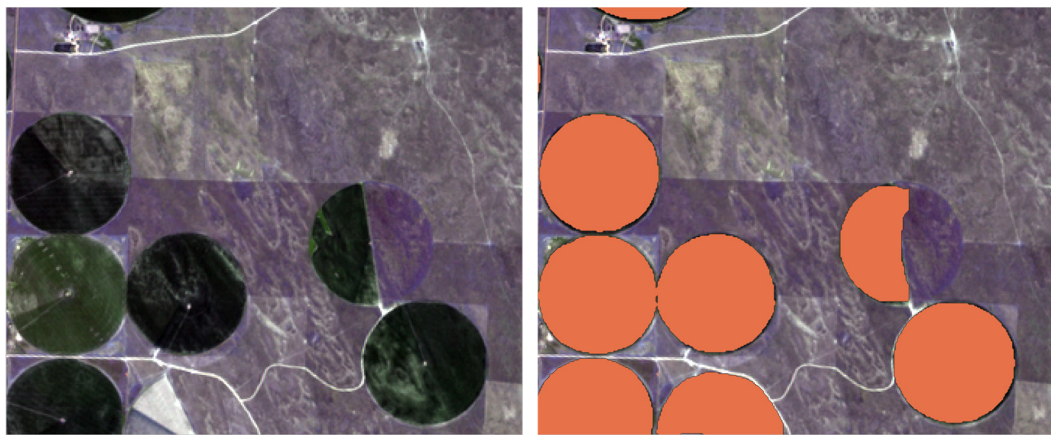


Fig. S10 Cities with a population of more than 500,000 and within 50 km of CPIS



(a)



(b)

Fig. S11 (a) Fully segmented results for partly vegetated areas; (b) partly segmented result for a partly vegetated area.

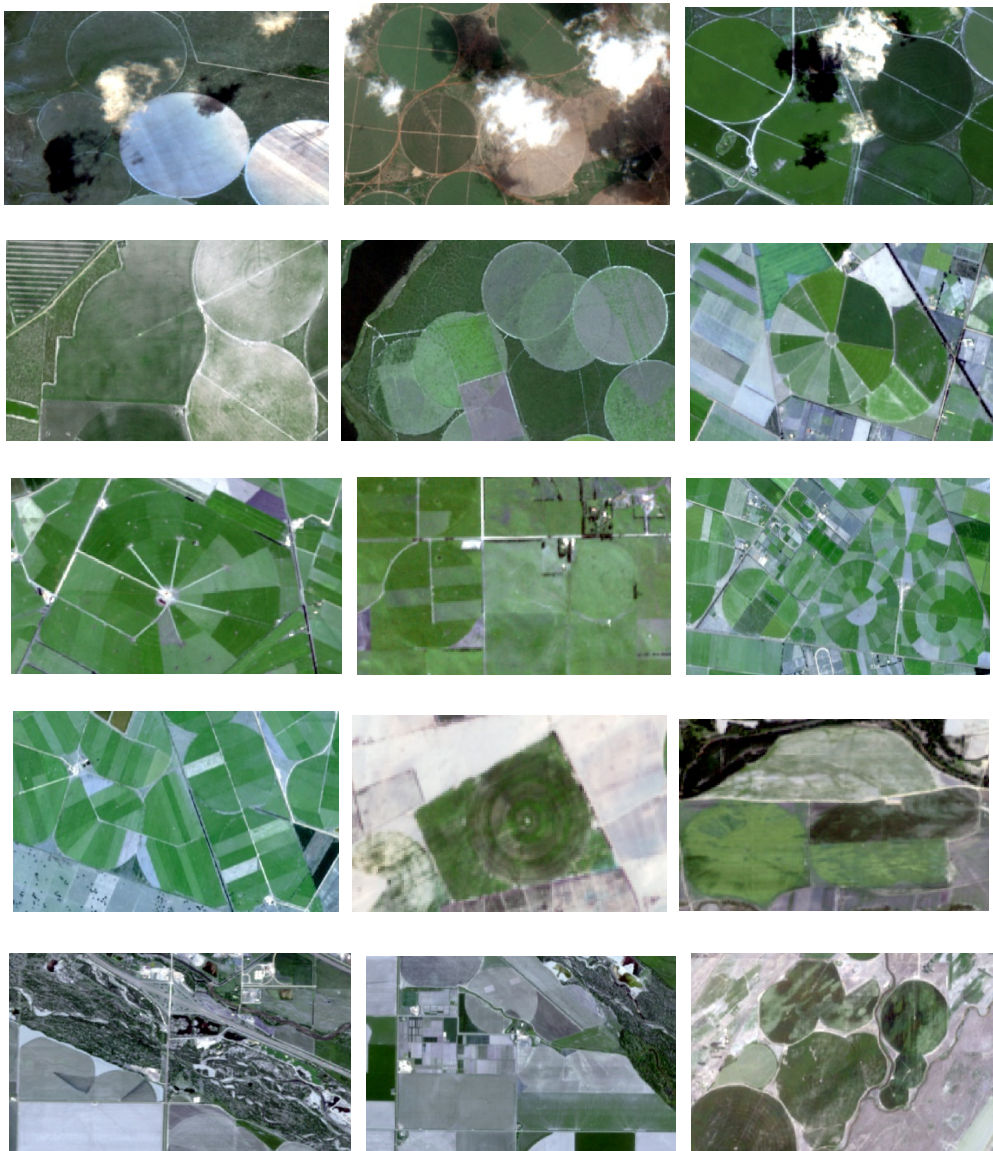


Fig. S12 Some example images of missed CPIS.